



Review

Old paradigms and new concepts in familial Mediterranean fever (FMF): an update 2023

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Abstract

Familial Mediterranean fever (FMF) is a hereditary autoinflammatory disease characterized by recurrent attacks of fever and polyserositis. Its first description as a new entity was published by Siegal in 1945. Colchicine has been the treatment of choice for this disease since 1972. Significant progress has been made over the years in understanding FMF's clinical features, diagnosis, mode of inheritance, pathogenesis and therapeutic approach. However, many old paradigms related to FMF have proven inaccurate, leading to the emergence of new concepts that provide more precise insights. The term 'FMF' is no longer appropriate as the disease is found beyond the Mediterranean basin. The concept of diagnosis based only upon clinical ground proved to be wrong. The paradigm that *MEFV* mutations in FMF lead to loss of function of the encoded peptide pyrin turned out to be a gain of function mutation. Finally, the concept that as a genetic disease FMF should be treated for life was found to be inaccurate for the subpopulation of the heterozygote patients. Thus, the breakthroughs of identifying the gene associated with the disease (*MEFV*) and the deciphering of its pathogenesis revolutionized our old paradigms and replaced them with new and more precise insights.

Keywords: FMF, *MEFV* gene, pyrin, colchicine, IL-1 blockers.

Rheumatology key messages

- FMF is an inaccurate name that should be replaced by pyrin-associated autoinflammatory disease (PAAD).
- FMF has various faces with diverse clinical features in different parts of the world.
- *MEFV* mutations increase the function of pyrin in its role of perpetuating inflammation (gain of function).

Introduction

Familial Mediterranean fever (FMF) is a hereditary autoinflammatory disease characterized by recurrent attacks of fever and polyserositis [1]. Each attack lasts about 24–72 h and abates spontaneously. The frequency of the attacks varies between patients and in the same patient in different periods of his life. Colchicine has been the treatment of choice since 1972 [2, 3]. It prevents the attacks of FMF and withholds the development of serum amyloid A (SAA), which is the most devastating complication of untreated FMF [1].

FMF was described as a new entity by Siegal in 1945 [4]. He named this disease 'paroxysmal benign peritonitis'. Over the years, the disease gained different names in order to better characterize its nature. The concepts about the genetics of FMF and its hereditary trait have also evolved over the years following the cloning of the *MEFV* gene associated with the disease [5, 6]. Similarly, our concept about the pathogenesis of the disease shifted from one side to another as we learned better the exact function of pyrin, the protein encoded by the *MEFV* gene. It was found that this peptide serves as a pivotal component in the inflammatory process underlying the acute

attacks of FMF [7]. Because the first descriptions of the clinical manifestations of FMF originated from Middle Eastern populations, our concept was that their presentation was the clinical prototype of FMF. However, when we encountered FMF patients from Japan and Europe we realized that the disease might display completely different clinical features. This observation changed our previous paradigm of the disease clinical characteristics [8]. Being a hereditary disease, it was assumed that FMF treatment with colchicine needs to be life-long. Recently, we discovered that this paradigm is no longer true in a sub-population of FMF patients who carry a single *MEFV* mutation [9, 10].

The objective of this review is to highlight the significant shifts in our old paradigms concerning the disease's name, clinical features, diagnosis, pathogenesis, and its therapeutic approach.

For this review, a comprehensive search was conducted through various databases from the first description of FMF in 1945 until June 2023, including PubMed, Embase, Scopus and MEDLINE. Data from old books and proceedings on FMF were also used [11, 12].

Shifting names of FMF

In 1908, Janeway and Mosenthal described a Jewish girl who had had recurrent attacks of fever and abdominal pain since infancy [13]. Each attack lasted 8–16 h and the frequency varied from 8 days to 1 month. They named the disease ‘toxemia of endogenous origin’ and it seemed to be the first documentation of FMF. In North Africa, recurrent bouts of fever have been mistaken for malaria; however, Mamou and Cattán were convinced that the disease was a real separate entity as early as 1925, although they published it much later [14]. It was only in 1945, however, that Siegal described the disease as a distinct clinical disorder which he named ‘benign paroxysmal peritonitis’ [4]. Three years later, Reimann, impressed by the recurrence of the attacks, introduced the term ‘periodic disease’ [15]. In 1957, Rachmilewitz *et al.* [16] proposed the name ‘recurrent polyserositis’, noticing that the target tissue during the acute attacks were serosal. Heller *et al.* [17] finally proposed the current name, ‘familial Mediterranean fever’, in the same year (1957). However, over the years, this name was found to be inaccurate because the disease may be present in countries outside of the Mediterranean basin (e.g. Armenia, Iran and Japan), sometimes it is not familial and, rarely, fever is not its main clinical feature. Therefore, in 2016 a committee was set in order to propose a new way of naming the recurrent fever syndromes [18]. Its aim was to look for names that will better characterize the clinical features or the pathogenesis of these diseases. Because the gene (*MEFV*) and the protein which it encodes (pyrin) were already known, the committee suggested the term pyrin-associated autoinflammatory disease (PAAD). This name includes all types of diseases associated with *MEFV* mutations (Fig. 1). Under this general ‘umbrella’ term, there are subtypes of disorders with different names such as periodic fever with autoinflammation and neutrophilic dermatosis (PAAND), chronic noninfectious osteomyelitis-like disease (CNO-like), FMF etc. Although, it is preferred to use the name of the gene over that of its encoded protein, in this case the name of the polypeptide—pyrin—was preferred. The reason is that the name *MEFV* was coined to denote its association with Mediterranean fever, which is no longer precise because its mutations may cause PAAND, CNO-like disease and livedoid reticularis, diseases totally different from FMF [18, 19]. Moreover, many of the diseases associated with the *MEFV* gene are not restricted to

Table 1. Designations used for familial Mediterranean fever

Name	Year	Reference
Toxemia of endogenous origin	1908	Janeway and Mosenthal [13]
Benign paroxysmal peritonitis	1945	Siegal [4]
Periodic disease	1948	Reimann [15]
La maladie periodique	1951	Mamou and Cattán [14]
Periodiche Fieber	1953	Hitzig [64]
Recurrent polyserositis	1957	Rachmilewitz [16]
Hyperpyretic pseudo infectious attacks	1957	Klotz [58]
Familial Mediterranean fever	1958	Heller <i>et al.</i> [17]
Familial recurrent polyserositis	1959	Priest [55]
Periodic peritonitis	1975	Bakir [62]

the Mediterranean Sea. Thus, the more accurate name is PAAD, which is an ‘umbrella’ term for all the diseases associated with abnormal pyrin. However, in this review we keep using the name FMF because it is still more popular among the readers. Table 1 lists some of the names proposed for FMF over the years.

Changing face of the disease

Based upon several series from the Middle East, a representative presentation of an FMF patient from this area includes: recurrent attacks of fever accompanied by serositis (peritonitis, pleuritis, pericarditis or synovitis) in one or more sites during the same attack. The classic skin manifestation of FMF is erysipelas-like erythema. The typical attack duration is 1–3 days [1].

Over the years, we have encountered many patients from around the world who received the diagnosis of FMF, though they presented with atypical clinical features [8]. For example, among Japanese patients with FMF, the frequency of fever and peritonitis is relatively low whereas complaints about headache are much more common [20, 21]. Moreover, the attack duration is longer (4–7 days) and the disease usually responds to a relatively low dose of colchicine compared with Middle Eastern patients with FMF. Among European FMF patients, many complained about oral aphthosis, erythematous pharyngitis, arthralgia, conjunctivitis, headache and enlarged cervical lymph nodes. In these cases, a question is raised regarding the accuracy of their diagnosis. The above clinical features among the European patients resemble better a diagnosis of periodic fever with oral aphthosis, pharyngitis and cervical adenopathy (PFAPA) rather than FMF [22].

In order to explain the differences between Japanese and European FMF patients and the Middle Eastern cases, one should look at the underlying genetics among the different FMF populations. Genetic analyses of Japanese patients disclose that only 26% of them carry *MEFV* exon 10 mutations [20, 21]. In these patients, the main mutation is M694I while none of them carries the M694V variant which is commonly seen among Middle Eastern patients with FMF. Additional exon 10 mutations such as M680I or V726A, which are frequently seen in the Middle East, are hardly ever found among Japanese patients. Seventy-four percent of Japanese patients carry mutations in exons 1–4 such as E148Q, P369S-R408Q (in Cis), L110P-E148Q (in Cis) and E84K. Some of them are

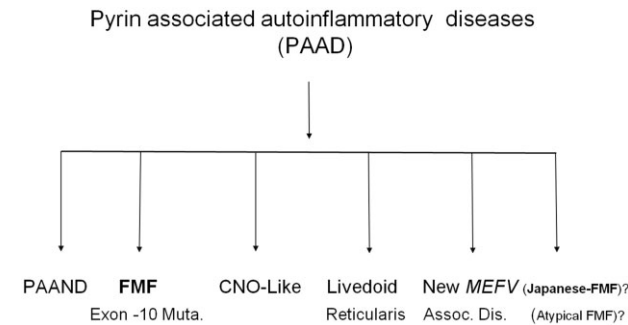


Figure 1. Names of diseases associated with *MEFV* mutations or variants. The ‘umbrella’ name is Pyrin Associated Autoinflammatory Diseases (PAAD) which is used as a general term. The subtypes are more specific and meaningful. PAAND: pyrin-associated autoinflammation with neutrophilic dermatosis; FMF: familial Mediterranean fever; CNO, chronic non-bacterial osteomyelitis. Livedoid reticularis was also related to *MEFV* mutations [19]

considered as polymorphisms rather than disease causing mutations. Nevertheless, in Japan they might cause a disease with clear genotype–phenotype correlation. Japanese patients carrying exon 10 mutations present with clinical features of Middle Eastern type of FMF, while those with other *MEFV* variants display atypical features. Based upon these observations, we postulated that exon10 mutations are responsible for the Middle Eastern prototype FMF, whereas those carrying non exon 10 mutations display variant FMF presentations [8].

Changes in diagnostic and classification criteria for FMF

Before the identification of the *MEFV* gene, several diagnostic and classification criteria have been established: the Tel Hashomer criteria, Livneh criteria and Yalcinkaya criteria for children with FMF [23–25]. These criteria did not include genetic testing and were based purely on clinical grounds. Some of them included response to colchicine treatment as an additional criterion. Recently, Karacan *et al.* [26] described two Turkish families in whom four patients presented with typical clinical features of FMF. Genetic analysis performed in these patients failed to show any *MEFV* mutation. However, whole exome sequencing (WES) revealed that two patients were homozygous for mutations in the *MVK* gene, which is associated with mevalonate kinase deficiency (MKD) and the other two patients carried mutations in the *TNFRSF1A* gene, which causes TNF receptor-associated periodic disease (TRAPS). These cases illustrate the difficulties in making a diagnosis of FMF based upon clinical features only and justify the need to add genetic testing for more precise diagnosis.

In a study by Gattorno *et al.* [27, 28], 360 cases with different autoinflammatory diseases—FMF, TRAPS, MKD, cryopyrin-associated periodic fever syndrome (CAPS) and PFAPA—were summarized and analysed. Their clinical features were presented to >20 clinicians and 10 geneticists for diagnosis. Then, their genetic results were added and the above clinicians were asked to revise (or not) their diagnoses based upon the addition of the new genetic data. Then, consensus (>80%) on diagnosis was reached in 281 patients and this served for creating two new sets of classification criteria: one purely clinically-based and the other included genetic testing. In the case of FMF, for the clinical criteria the authors proposed nine items of which a patient with recurrent episodes of fever and elevated inflammatory biomarkers has to fulfill six. The items contained are: Eastern Mediterranean ethnicity, attack duration of 1–3 days, chest pain, abdominal pain or arthritis. In addition, lacking one or more of the following items also allowed the patient one point: (for each of them) aphthous stomatitis, urticarial rash, maculo-papular rash or painful lymph nodes. Thus, if a patient gains at least six points, they may be classified as having FMF with specificity and sensitivity of 92% and 91%, respectively. For the genetic set of criteria, the authors used a pathogenicity score for each *MEFV* variant [29]. In addition, they defined another term ‘confirmatory *MEFV* genotype’ where a patient carries pathogenic or likely pathogenic mutations. In case a patient with recurrent episodes of fever has a confirmatory *MEFV* genotype, they should fulfill at least one of the following items in order to have a confirmed diagnosis of FMF: attack duration of 1–3 days, arthritis, chest pain or abdominal pain. Alternatively, in the presence of non-confirmatory *MEFV*

genotype (carriage of a variant of uncertain significance or most likely a benign mutation), the patient should have two of the above items in order to be classified as having FMF. A major limitation of these criteria is the need for the treating physicians to be familiar with the area of FMF and autoinflammatory diseases.

In summary, we see that following the identification of the gene associated with FMF (*MEFV*) we refined our diagnostic and classification criteria for FMF. This may help us better understanding the prognosis of the patients and their appropriate treatment. Those with confirmatory genotype usually have more severe disease which may require a higher dose of colchicine. In addition, we learn that for a definitive diagnosis of FMF, we need genetic testing results, whereas for probable diagnosis and for a decision to try colchicine treatment, we may rely upon clinical criteria only.

Recently, two new tests were proposed for the diagnosis of FMF. One uses an *ex vivo* colchicine functional assay in peripheral blood mononuclear cells in patients and controls [30]. The other assay quantifies the effect of UCN-01 (a kinase inhibitor) on inflammasome activation, thereby distinguishing FMF patients from healthy donors [31]. An inflammasome is a cytosolic multiprotein complex, classically consisting of an upstream sensor protein of the NOD-like receptor (NLR) family or pyrin, the adaptor protein ASC, and the downstream effector procaspase. The main limitation of these tests is the need for quite sophisticated methods and well-equipped laboratories, which are not commonly found in most medical centers. Moreover, they would not eliminate the need for genetic testing.

Changing concepts of FMF pathogenesis

Over many years, we did not understand the exact pathogenesis of FMF. Following the identification of the *MEFV* gene and its encoded protein (pyrin), the primary concept was that pyrin attenuates the inflammatory process. On the other hand, the *MEFV* mutations were supposed to cause loss of pyrin’s suppressing function, thereby enhancing inflammation [32] (Fig. 1). Papin *et al.* [33], showed that pyrin C-terminal domain interacts with caspase-1 and its substrate pro-IL-1b. They demonstrated that whereas a pyrin knockdown results in increased caspase-1 activation and IL-1b secretion, overexpression of the pyrin C-terminal domain alone blocks these processes. Chae *et al.* [34] demonstrated that the C-terminal B30.2 domain of pyrin is necessary and sufficient for the interaction with the caspase-1 and this binding was reduced by FMF-associated B30.2 mutations. Full-length pyrin attenuated IL-1 production in cells transfected with a caspase-1 IL-1 construct, an effect diminished by FMF-associated B30.2 mutations and in B30.2 deletion mutants. Thus, pyrin encoded by mutated *MEFV* gene further loses its function as a suppressor of inflammation underlying the acute attack of FMF (Fig. 2). In contrast, the group of Alnemri *et al.*, claimed from the beginning that the pyrin protein serves as a pro-inflammatory agent rather than an inhibitor of inflammation [35, 36]. They demonstrated that PSTPIP1 peptide requires pyrin in order to assemble the ASC pyroptosome (apoptosis-associated speck-like protein containing a caspase recruitment domain – an adaptor peptide in creating the inflammasomes), a molecular platform that recruits and activates caspase-1. Pyrin exists as a homotrimer in an autoinhibited state due to intramolecular interactions between its pyrin domain (PYD)

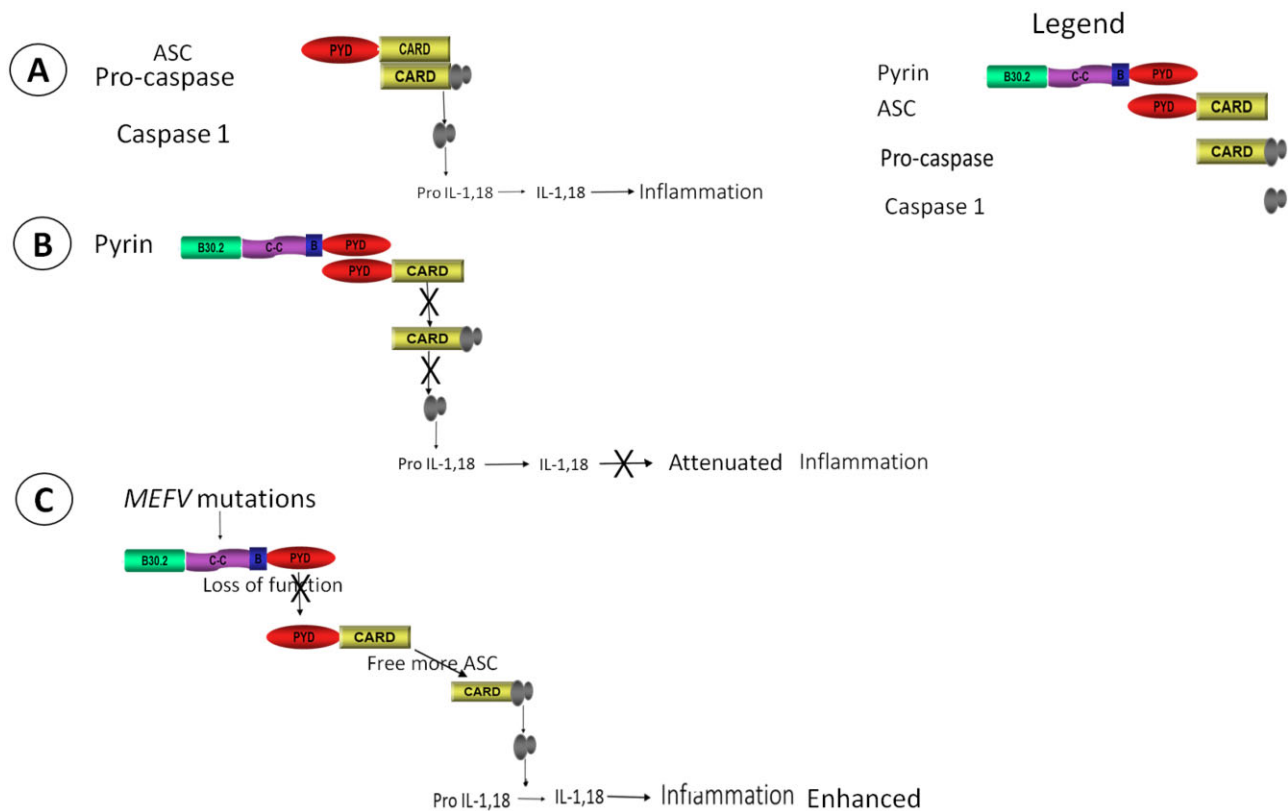


Figure 2. A schematic illustration of the pyrin action as an attenuating factor of inflammation. **(A)** Following an interaction with an external or internal danger signal the ASC (apoptosis-associated speck-like protein containing a caspase recruitment domain) recruits procaspase through its CARD domain (caspase activation and recruitment domain) (CARD–CARD interaction) which leads to activation of caspase 1. Active caspase 1 converts pro-IL-1 and pro-IL-18 into an active form of IL-1 and IL-18 which perpetuate the inflammatory process. **(B)** Following an interaction with a false or true danger signal, the pyrin protein recruits the ASC through its pyrin domain (pyrin–pyrin interaction). The ASC is trapped by the pyrin protein and therefore can no longer interact with procaspase so that the process of inflammation is halted. **(C)** Mutations in the *MEFV* gene encode pyrin, which has reduced affinity and capability of interaction with ASC, thereby allowing it to bind freely procaspase and to enhance the inflammatory process. Thus, *MEFV* mutations cause loss of pyrin function

and its B-box portion. Ligation by PSTPIP1, which is also a homotrimer, activates pyrin by unmasking its pyrin domain (PYD), thereby allowing it to interact with ASC and facilitate ASC oligomerization into an active ASC pyroptosome. Human pyrin with mutations in its B30.2 domain enhances the assembling of an ASC-dependent inflammasome that activates caspase-1.

In 2014, Xu *et al.* [37] discovered that pyrin is a pattern recognition particle (PRR) which can assemble ASC and procaspase to create the ‘pyrin inflammasome’. This occurred following exposure to clostridium toxin. Their observation was that clostridium toxin could reduce the effect of RhoA protein on GTPases (in peptides phosphorylation), thereby increasing pyrin activity followed by higher production of IL-1. However, there was no explanation for the missing link between the reduction of RhoA effect and the increased activity of pyrin. Two groups of investigators cracked the exact mechanism of pyrin action and filled this gap [38, 39]. It was known already that phosphorylated pyrin is bound by the inhibitory 14–3–3 proteins and this process retains pyrin in its inactive state that prevents formation of a pyrin inflammasome [40] (Fig. 3). However, when pyrin is not phosphorylated due to an external cause (such as an infection, etc.) or due to an internal cause (mutations, which affect the site of phosphorylation), it is not bound by the 14–3–3 peptides and

therefore, a high amount of non-phosphorylated active pyrin remains free for the construction of the pyrin inflammasome, thereby perpetuating the inflammatory process [39]. Thus, we see that mutated *MEFV* gene leads to a gain of function of the pyrin towards more inflammation.

The above observations changed our concept of the pathogenesis of FMF by 180 degrees over the last few years, due to our better understanding of the pyrin function. Another paradigm related to FMF turned over.

Changing mode of inheritance

When FMF was defined as a new entity, two basic observations supported an autosomal recessive inheritance: first, because in the vast majority of affected families the disease occurred in members of only one generation and second, because the high prevalence of consanguinity could explain the occurrence of the disease in two or more successive generations.

Following the identification of the gene associated with FMF (*MEFV*), it was found that in the Middle East, two-thirds of the patients carry two mutations and about a third of FMF patients have a single mutation [41, 42]. The observation that the majority of patients carry two mutations also supports the notion of autosomal recessive inheritance. Still,

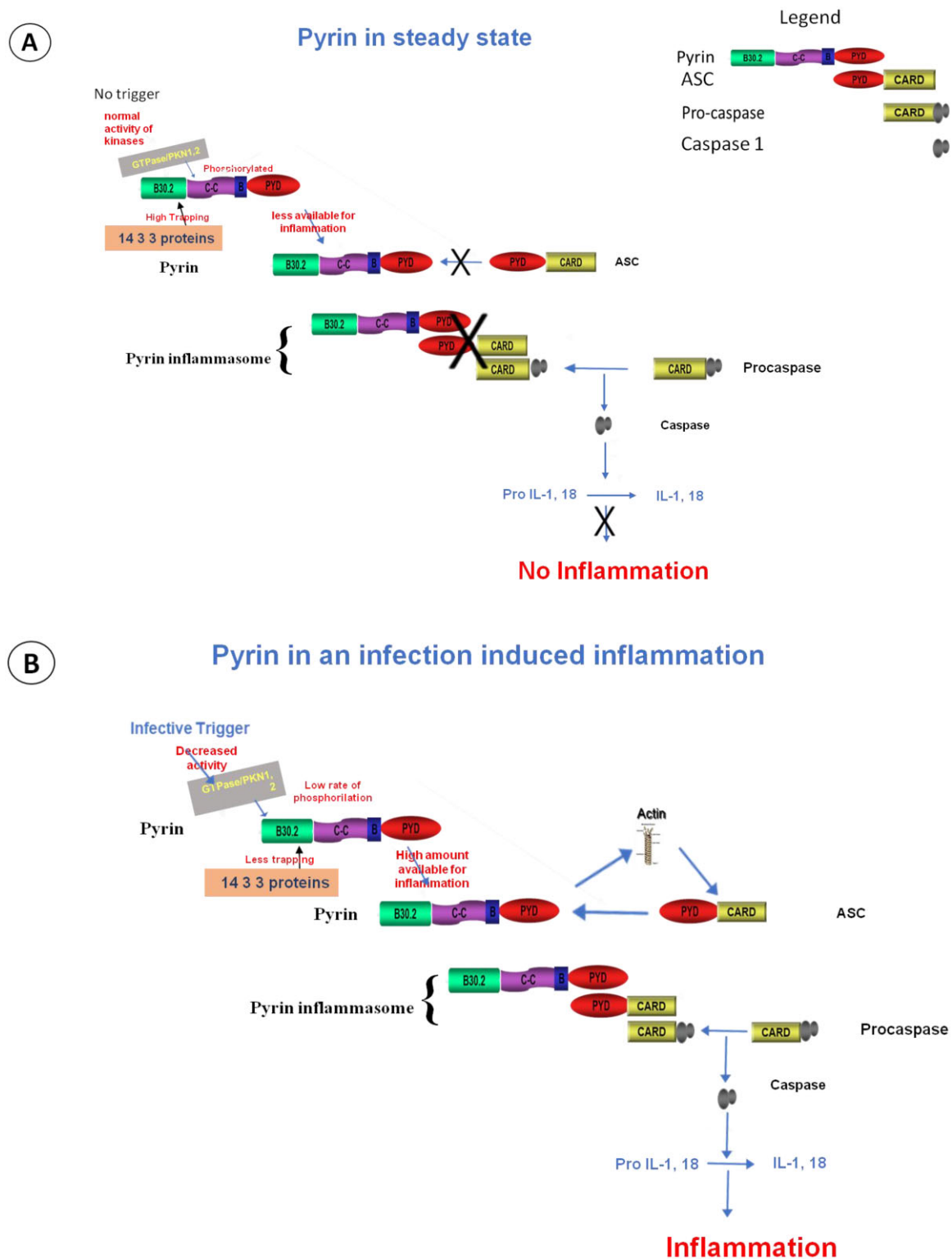


Figure 3. A schematic illustration of the pyrin action as an enhancing factor of inflammation. **(A)** In a steady state, pyrin is phosphorylated by protein kinases. Phosphorylated pyrin is trapped by 14–3–3 peptides and therefore it can no longer participate in the process of inflammation through an interaction with ASC and procaspase for the construction of the pyrin inflammasome. Without the inflammasome there is no inflammation. **(B)** Following an interaction with an external or internal danger signal (such as infection), the rate of pyrin phosphorylation by protein kinases is reduced. More non-phosphorylated pyrin remains because it is not trapped by the 14–3–3 proteins. This active non-phosphorylated pyrin recruits ASC and procaspase in order to create the pyrin inflammasome. Following the construction of the pyrin inflammasome, the procaspase releases an active caspase which perpetuates the inflammatory process by converting pro-IL-1, 18 to active IL-1, 18. **(C)** In FMF patients, mutations in the *MEFV* gene further reduce the phosphorylation of pyrin and increase its affinity to ASC without the need for actin mediation, thereby enhancing the direct construction of the pyrin inflammasome (by immediate recruitment of ASC and procaspase towards bursting inflammation and FMF attack)

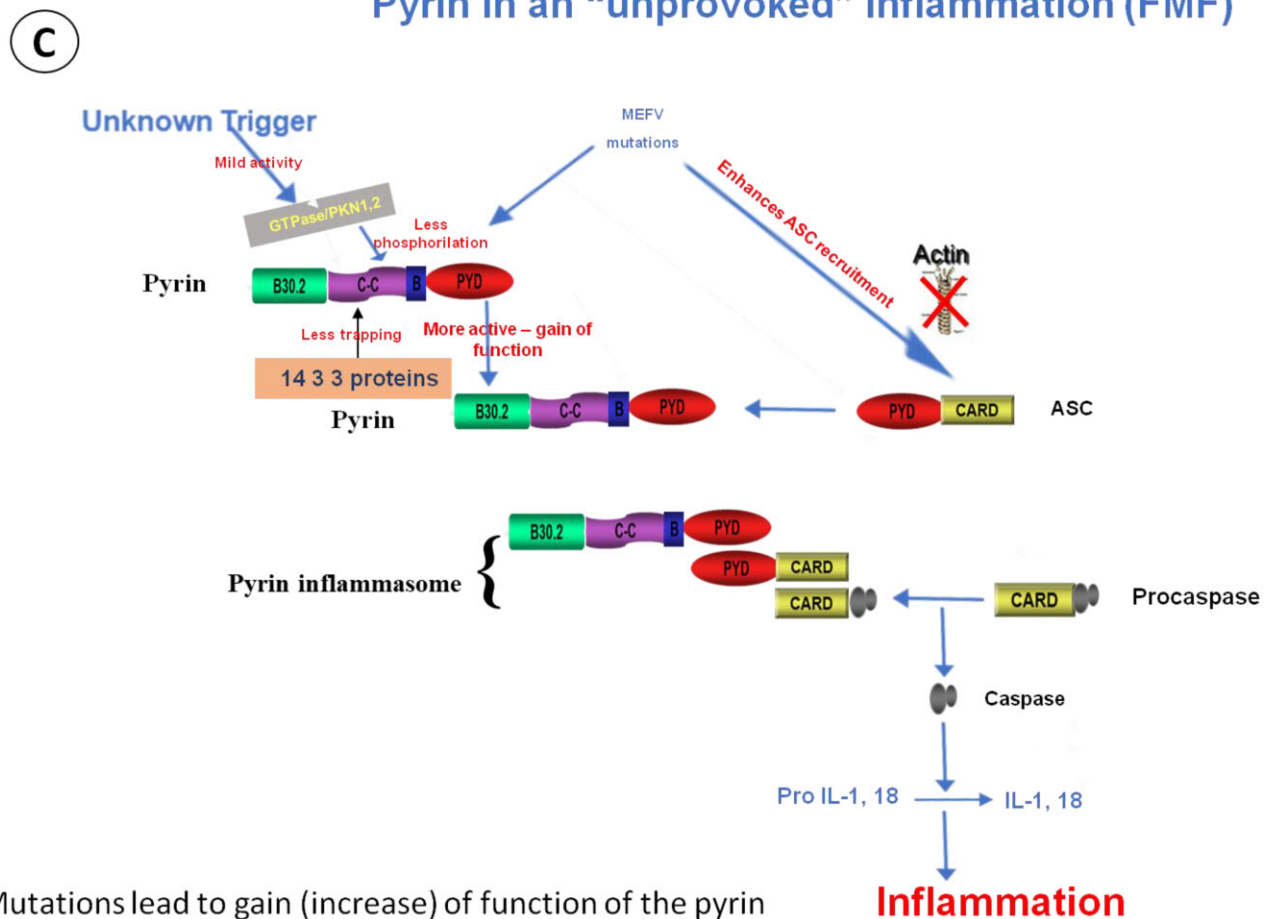


Figure 3. (Continued)

the fact that a third of the patients carry a single mutation called for an explanation. Indeed, numerous studies looked for additional genetic modifiers among these patients; none of which succeeded in finding such a contributive factor [43, 44].

Another support for the recessive mode of inheritance comes from the observation that >95% of the heterozygotes found in the general population screening for *MEFV* variants are asymptomatic [45]. In a study by Jéru *et al.* [45], they suggested that the carriage of a single mutation is a susceptible condition for developing disease but not necessarily a cause for it. According to their calculation, carrying a single *MEFV* mutation increases the risk for FMF by 6–8-fold compared with healthy individuals who do not carry it.

Although the evidence for recessive autosomal mode of transmission is overwhelming, claims that transmission is dominant have also been made. In a three-generation Spanish kindred, five affected members presented a severe periodic inflammatory disorder segregating with the rare p. H478Y *MEFV* variant located in exon 5 [46]. These patients had longer attack duration, resistance to colchicine and a dominant inheritance in contrast to the Middle Eastern type of FMF. In a family of British descent with a colchicine responsive periodic fever syndrome, WES revealed a missense mutation in exon 8 p.T577N [47]. Their disorder was similar to FMF (fever, systemic inflammation, a response to colchicine or IL-1 blockade) but somewhat distinct from it

(dominant inheritance, longer duration of attacks and evanescent urticaria-like skin rash). Dominant inheritance of the same genetic variants (p.T577N) was also observed in a Japanese family with PAAD [48].

The above findings support the notion that FMF is usually inherited as an autosomal recessive trait. However, in some cases with specific *MEFV* mutations the transmission may be dominant. Thus, our previous paradigm of typical recessive mode of inheritance has also been changed over the years.

Screening Middle Eastern populations reveals that many asymptomatic individuals carry a single pathogenic variant [49]. This raises a theoretical possibility that these persons have a selective advantage. Over many years a hypothesis was proposed that the heterozygotes better survived an ancient infective agent which cannot be found nowadays. Recently, Chung *et al.* [50] showed that *Yersinia pestis* and pseudotuberculosis bacteria can downregulate pyrin activity. *Yersinia* Outer Protein (YOP) M can recruit protein kinase 1 and 2 and enhance the phosphorylation of pyrin. This in turn, increases its trapping by the 14–3–3 peptides so that less available active pyrin remains for the construction of the pyrin inflammasome towards producing more IL-1 and IL-18. In heterozygotes, the carriage of a single mutation reduces the rate of pyrin phosphorylation, thereby increasing the amount of active pyrin for propagating the inflammatory process in order to fight the invasive pathogen. Thus, it may well be that heterozygotes have a better resistance and survival

in times of significant infections and plagues. This observation represents a case where a theoretical hypothesis about selective advantage found its impressive support in real-life and laboratory evidence.

Changing concepts in treating FMF

At the time when FMF was first described, there was no treatment for the disease. The whole gamut of sulfa drugs and antibiotics, as they became available, were tried in FMF patients [51–55]. It was easy to establish that none of them was effective in the prevention of attacks. At one time or another all the following therapeutic measures were more or less enthusiastically employed by some and said to be of no value by others: ACTH and corticosteroids [17, 56], estrogen, testosterone [55, 57–60], salicylates [17, 53, 54], phenylbutazone [53, 54], indomethacin [61] quinine or quinidine [17, 61] and antihistamines [62, 63]. Anticonvulsive agents [59, 64], arsenic compound [17, 65], cholestyramine [61], reserpine [66] lithium [67], paravertebral block and sympathectomy [15] all have been tried unsuccessfully.

In 1972 Goldfinger and Ozkan introduced colchicine for the treatment of FMF and showed that it was effective in preventing the acute attacks of the disease [2, 3]. The efficacy of colchicine in FMF was observed in patients who had concomitant gout for which this medication was given [2]. Later, Zemer *et al.* [68] showed that colchicine was also effective in halting the development of amyloidosis. Since the introduction of colchicine for the treatment of FMF, all the above listed drugs have practically become of historical interest.

Over the years, we studied the pharmacokinetics and pharmacodynamics of colchicine and learned how to use it properly [69, 70]. We learned its side effects and its high potential of toxicity in cases with renal or liver failure or when it is administered concomitantly with other medications [71]. We also investigated its safety in pregnant and nursing women with FMF and in children [72–74]. Recently, EULAR published therapeutic guidelines for FMF and as a whole, the quality of life and the longevity of FMF patients have improved significantly [75]. However, about 5–10% of FMF patients do not respond to colchicine (resistants) and about 15% are intolerant to the drug even in its low dose [76, 77]. This high rate of intolerance was recently shown in a study which tested the effectiveness of colchicine (0.5 mg daily) in a secondary prevention of cardiovascular disease [77]. This sub-group of FMF patients who are resistant or intolerant to colchicine pose a serious challenge to treatment. Following the identification of the *MEFV* gene and the understanding of the FMF pathogenesis, it was found that IL-1 plays a pivotal role in the inflammatory process of the disease [39]. This observation led us to try anti IL-1 agents in FMF patients, which was proven very successful [78–80]. Thus, today we do have a reasonable solution for patients with FMF who are resistant to colchicine.

Anecdotal reports suggested that anti-IL-1 agents could also prevent or slow the development of amyloidosis in FMF patients [81–84]. However, a small study by Simsek *et al.* [85] failed to show a similar success. Yet, it should be emphasized that adjusting the doses of IL-1 inhibitors to effectively suppress inflammation is crucial, particularly in individuals already having amyloidosis. Failure to tailor the doses may result in poor long-term prognosis [86]. Up until now there have been no controlled trials demonstrating the efficacy of anti-IL-1 medications in preventing amyloidosis. Therefore, our policy is still to give colchicine (1–1.5 mg daily)

concomitantly with the anti-IL-1 agents in order to prevent the development of amyloidosis [87].

Another point of a shifting paradigm relates to our treatment approach of FMF patients carrying a single mutation (heterozygotes). In the past, the concept was that because FMF is a genetic disease, we should treat the patient with colchicine for life. In recent years, we observed that there are heterozygotes FMF patients in whom we can discontinue colchicine without any negative outcomes [9, 10, 88–90]. Patients who are candidates for such an approach should carry a single *MEFV* mutation and should be free of FMF attacks for at least 4–5 years (unpublished data). In such patients, we can stop colchicine and follow them in the clinic for FMF attacks and monitor their inflammatory parameters. We should also check their urine for amyloidosis-induced proteinuria, which is a relatively rare complication among heterozygotes.

Thus, we see that since 1972 there has been enormous progress in our treatment approach of FMF patients. We hardly see today any patients who develop amyloidosis and renal failure. Owing to our current treatment with colchicine and anti-IL-1 agents, the prognosis of FMF patients has become almost comparable to that of the general healthy population.

Summary

A review of FMF disease over the years has disclosed that many previous concepts and paradigms related to this disease have been changed. This includes the aspects of naming FMF, defining its typical clinical features, establishing its classification criteria, pathogenesis, genetics and the therapeutic approach. These great changes were the results of several breakthroughs which took place in the field of FMF since its first description. The first important step forward was the introduction of colchicine for treating FMF in 1972 [2, 3]. The next significant progress was the identification and cloning of the gene associated with the disease (*MEFV*) [5, 6]. The next breakthrough was the understanding of the exact role of pyrin, as a pattern recognition particle and as a pivotal component of the pyrin inflammasome [37, 38]. Finally, the cracking of the exact mechanism of pyrin inflammatory action paved the way for the use of anti-IL-1 agents in FMF patients who are resistant or intolerant to colchicine [39].

Several important points have been overlooked in this review. These include the predictors for and the impact of chronic inflammation in FMF [91–93], co-morbidities associated with this disease [94–97] and the issue of long-term prognosis of FMF [98–100]. The reason for that is the limitation of word space in the current review. These important items merit future consideration and elaboration. Additional potential directions for future research may include the following topics:

- 1) Why do FMF attacks last 24 to 74 h and resolve spontaneously?
- 2) How do we explain the fact that many FMF patients have a single mutation whereas many heterozygotes remain asymptomatic?
- 3) What is the explanation for the observation that despite colchicine inhibition of neutrophils chemotaxis, FMF patients do not experience more infections?

In conclusion, it seems that the impressive progress made in FMF is one of the most significant achievements in medicine

in the past years. It led to a revolutionary improvement in controlling FMF attacks and improving the quality of life of the patients. In addition, FMF has become a pioneer syndrome which underlies the discovery of the new large family of autoinflammatory diseases [101].

Data availability

The data underlying this article are available in the article.

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